

“The organic cell technology is at a very nascent stage and it will take some time to replace the current single-crystal silicon solar energy solutions — Dr V.K. Kaul, general manager, Solar Photovoltaic Group, Central Electronics Ltd (CEL)

into substantial difference in the power output towards the end of the project.

“Thin-film technology is constantly being improved to overcome the stability and degradation issues and many manufacturers now provide warranties similar to those provided for crystalline modules. Although commercially available thin-film modules have lower rated efficiencies than crystalline modules, these often perform better in partially shaded conditions and have a lower temperature coefficient (allowing their performance to degrade less in high ambient temperature conditions),” shares Choudhury.

Within thin films, amorphous silicon is losing ground to cadmium telluride (CdTe). CIGS (copper indium gallium di-selenide) is also making inroads.

The latest breakthrough in cell technology is the introduction of dye or organic cells. Organic solar cells are flexible and lightweight, and their manufacturing cost is about one-third that of an inorganic solar cell. However, the main drawback of organic cells is their lower efficiency in converting sunlight into electricity. In general, organic cells have a 1-3 per cent efficiency rate, compared to silicon-based solar cells’ 15-20 per cent. Still, the organic cells technology scores over conventional solar cells because of its low impact on the environment and easy manufacturing process. Also, because these cells can be attached to flexible materials, they can be put on many surfaces.

“This technology is at a very nascent stage and it will take some time to replace the current single-

crystal silicon solar energy solutions. Rather, they are seen as an opportunity in their own niche market. As time progresses and the focus of the research concerning these cells broadens, the dye/organic solar cell might just become a contender in the current solar panel market,” opines Dr V.K. Kaul, general manager, Solar Photovoltaic Group, Central Electronics Ltd (CEL).

“Eventually, the technology which can deliver the maximum power output over the 25-year lifecycle of the project at the most competitive cost and without any environmental hazard will survive and thrive,” says Subramanya.

Can we meet the demand?

The solar capacity in India has far exceeded the demand in last ten years including 2011. The country still continues to be a large exporter of solar PV products. However, now with the NSM and state policies driving demand in the domestic market, the industry is well geared to meet this demand as well.

“The key drivers for solar power demand in India are a favourable and stable policy framework, support for solar-based generation, rising energy and fuel cost, declining solar system prices, removal of administrative

and bureaucratic hurdles, and availability of financing for solar projects at preferential interest rates,” says Surapaneni.

The biggest demand driver, however, is the sheer demand-supply gap in the overall power sector in India which stands at 12 per cent and peaks at 16-17 per cent. This is when 40 per cent of the country’s 1.2 billion population—that’s half a billion people—is not even connected to electricity!

The conventional model of large power plants with the grid trying to connect every nook and corner of the country has failed since the “pipe runs dry” by the time it reaches the last mile in the remote villages. Therefore decentralised renewable energy systems are essential to generate power where it is needed and consume it on the spot. Solar is an appropriate answer to address the energy security requirement of India.

Another demand driver for solar power is the emphasis on moving away from fossil fuels in order to meet our CO₂ emissions reduction commitment. As much as 54 per cent of the power generation capacity in India is coal-based and this accounts for 61 per cent of India’s GHG emissions. A shift away from coal over the long term is a national goal but this cannot happen until an efficient, accessible and affordable alternative is available. Solar has the potential to fit into this slot over the long term.

Challenges remain

The biggest challenge is the reluctance of banks and financial institutions (FIs) to finance the solar power projects. Projects which cannot achieve financ-



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